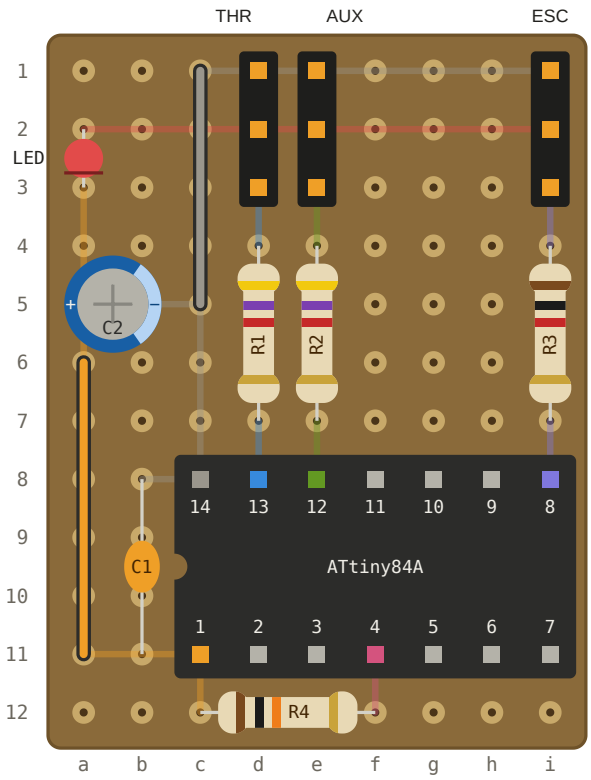
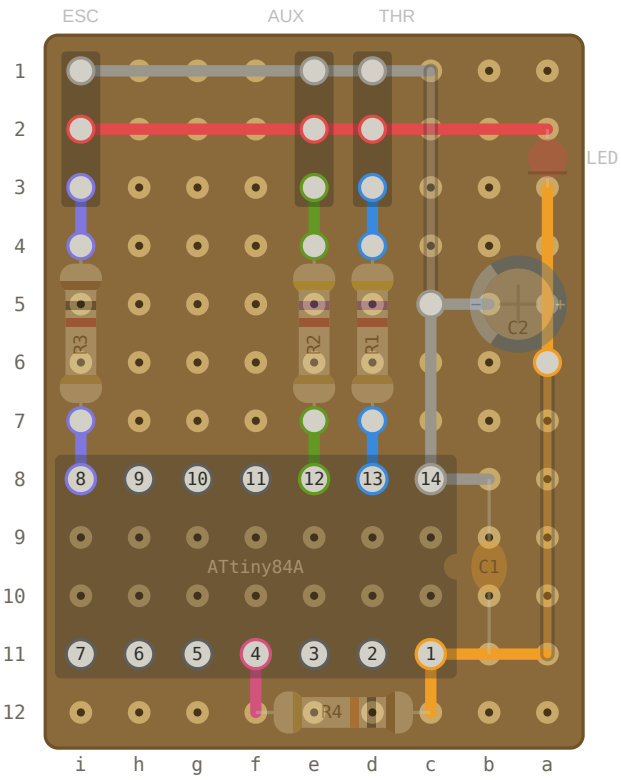
ATinyCrawl · Perfboard

ATtiny84A · 9 × 12 holes · approx. 23 × 30 mm · BEC 6 V

COMPONENT SIDE

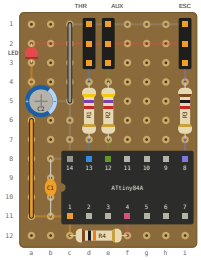


SOLDER SIDE (MIRRORED)

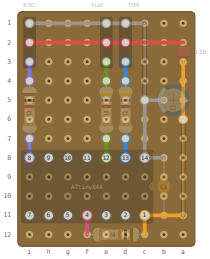


GND · +6V · VCC · CH2 · CH3/4 · ESC · RST

Component side: faint lines = solder bridges on the back, thick outlined lines = insulated wires on top. Solder side: the way the board lies in front of you while soldering, parts faint in the background.



Component side 1:1



Solder side 1:1

Scale 1:1 when printed at 100 % (2.54 mm grid).
Hold it against the perfboard.

Parts and checks

ATinyCrawl · RC Speed Limiter · ATtiny84A

PARTS

U1	ATtiny84A in 14-pin socket, c8–i11, notch on the left **Also ok:** ATtiny84 (without A). ATtiny44A or 24A should work too (the sketch needs 1.4 KB flash and 35 bytes RAM; builds, not tested on hardware). Select the chip accordingly. Soldering it in without a socket works, but then it can't be reflashed
LED	red, anode in a2, leg bent along row 2 on the back to d2 and soldered there; cathode in a3, leg bent on the back to a6 **Also ok (6 V BEC):** any 3 or 5 mm LED with 1.7–2.2 V forward voltage: red, orange, yellow, classic green. Not white, blue or bright true-green (≈3 V, too little left for the ATtiny). Other BECs: see table below
C1	100 nF ceramic, legs through b8 / b11, bent on the back to pin 14 / pin 1 and soldered only there **Also ok:** any ceramic 47 nF–1 µF (marked 473, 104, 224, 474, 105)
C2	10–100 µF electrolytic, 6.3 V or more, standing, + a5, − b5. Legs bent on the back to a6 and c5, no solder on a5 and b5 **Also ok:** 4.7–220 µF, any voltage rating ≥ 6.3 V; a 10 µF ceramic or tantalum works too (tantalum: mind the polarity)
R1, R2	4k7, d4–d7 and e4–e7 4-band 5-band yellow · violet · red · gold | yellow · violet · black · brown · brown **Also ok:** anything 1 k–10 k (e.g. 2k2, 3k3, 10 k), R1 and R2 don't even have to match
R3	1 k, i4–i7 (protects pin 8 if a plug is reversed) 4-band 5-band brown · black · red · gold | brown · black · black · brown · brown **Also ok:** 330 Ω–2k2 (e.g. 470, 680, 2k2). Below 330 Ω the protection gets weak
R4	10 k reset pull-up, c12–f12, legs bent on the back to pin 1 / pin 4 and soldered only there 4-band 5-band brown · black · orange · gold | brown · black · black · red · brown **Also ok:** 4k7–47 k
J1–J3	3-pin headers, columns d / e / i, − at the top **Also ok:** any 2.54 mm male pin header, cut to 3 pins (straight or angled)
Rails	bare wire on the back, row 1 (GND) and row 2 (+6V), each from d to i

WIRES ON TOP (INSULATED)

- GND: c5 → c1 on top. At c1 through to the back, bent to d1. At c5 through to the back, bent to pin 14 (c8)
- VCC: a6 → a11 on top, through a11 to the back and bent to pin 1 (c11)

CHECK BEFORE INSERTING THE CHIP

- No solder on a2–a5, LED soldered only at d2 and a6
- d2 ↔ a6: no continuity (only through the LED)
- a6 ↔ c5 and pin 1 ↔ pin 14: no continuity
- No solder on b5 (C2 soldered only at a6 and c5)
- +6V ↔ pin 1: no continuity
- With chip, powered: pin 1 ↔ 14 approx. 3.8–4.5 V

OTHER BEC VOLTAGES

5 V	1N4148 instead of the LED, band towards a3. Also ok: 1N4001–1N4007, 1N5817/1N5819, or a wire bridge (no power indicator then)
6 V	red or yellow LED. Also ok: orange or classic green LED
7.4 V	white or blue LED, measure to confirm